

PRAKHAR
SHEKHAR
PARTHASARTHI

12-INCH LP
LONG PLAY · 26

CAT. № PSP-26
33⅓ RPM · STEREO

LINER NOTES BY P. SHEKHARENGINEER · ARRANGER · PRODUCER

PRAKHAR.MNNIT.2022@GMAIL.COM · GITHUB.COM/PRAXSTACK · LINKEDIN.COM/IN/PRAKHARSHEKHAR · PRAX-
PORTFOLIO-ONE.VERCEL.APP · SHAHJAHANPUR, UP · OPEN TO RELOCATE

33⅓ RPM



STEREO

P90 LATENCYTRK · A1

-47%

Search fan-out · 1.10s → 0.59s, sustained across 3 quarters.

CONVERSIONTRK · A2

+33%

Traveler funnel uplift · measured in-prod.

CPTTRK · A3

-43 BPS

Payments cost-per-txn · idempotent webhook.

ON-CALLTRK · A4

~5 / SHIFT

From 20+ · alarm hygiene & runbooks.

SIDE A

THE MAIN FEATURE · TRACK LISTING

AMAZON TRAVEL · JUL 2023 — SEP 2025

SDE II · AMAZON TRAVELJUL 2023 — SEP 2025
Hotels · Flights · Insurance — Owner: 4 Java microservices, search → webhookTENURE · 2Y 2MBENGALURU, IN

Recorded live on the critical path from first pricing fan-out to post-booking webhook reconciliation. Four tracks, one concept album — re-cut concurrency, harden idempotency, re-paving pipelines, and a quieter pager. Mixed & mastered in production.

- A1 [05:47] CompletableFuture.allOf fan-out with bulkhead ExecutorService pools, replacing serial supplier calls — p90 1.10s → 0.59s (-47%) · traveler conversion +33% across 3 quarters.
- A2 [03:33] Idempotent payment webhook on SQS + Step Functions with a DynamoDB dedupe table keyed by correlation id — killed the double-capture SEV class, drove CPT -43 bps.
- A3 [04:43] Rewrote 6 legacy deploy pipelines into Java CDK with typed constructs & contract tests — rollbacks halved, MTTR -1 day, config drift traced to a single source of truth.
- A4 [05:20] On-call from 20+ → ~5 pages/shift, config-hygiene 5-6 → 0, resolved 150+ incidents, stopped an est. ~30 bps funnel-churn on insurance attach.

SIDE B

DEEP CUTS & LINER NOTES

EARLY SIDES · MAY 2021 — JUL 2023

B1 · SDE I · AMAZON TRAVEL

JUL 2022 — JUL 2023

TENURE · 1Y 0M

Bengaluru, IN — Services, DAOs, and a 60% renewal lift

Side B opens with the session work: **REST** services and **@Repository DynamoDB DAOs** wired up via Spring DI, then shipped the auto-insurance renewal flow with retry + idempotency — +60% renewal completion. Raised the test bar to 80%+ JUnit coverage (JaCoCo, CI-gated).

B2 · SDE INTERN · AMAZON TRAVEL

MAY 2021 — FEB 2022

TENURE · 10M

Bengaluru, IN — GraphQL → Java migration & One-Click Renewal

The demo tape: led a **GraphQL → Java** migration for a slice of the insurance quote path, and built **One-Click Renewal** on **Guice DI** with per-module factories. Coverage landed at 90%+ — converted to FTE on the back of it.

INSTRUMENTS

PERSONNEL · STACK CATALOGUE

P. SHEKHAR · 8 SESSIONS

JVM · CORE	P. Shekhar · Java 17, Spring Boot, Spring MVC, Spring DI, Guice, Maven, Gradle	CONCURRENCY	P. Shekhar · ExecutorService, CompletableFuture, ConcurrentHashMap, ScheduledExecutor, bulkheads
AWS	P. Shekhar · Lambda, DynamoDB, SQS, Step Functions, API Gateway, AppSync, CloudWatch, ECS, CDK (Java)	QUALITY	P. Shekhar · JUnit 5, Mockito, JaCoCo, integration harnesses, contract tests
FRONT-END	P. Shekhar · React (SSR & CSR), Redux, TypeScript, Vitest	DATA PATTERNS	P. Shekhar · DynamoDB single-table, idempotency keys, dedupe queues
OPS	P. Shekhar · Runbooks, alarm hygiene, SEV pages, on-call rotations	METHOD	P. Shekhar · Trunk-based dev, code-review, postmortems, SLO-first

BONUS

B-SIDES & BONUS TRACKS · PROJECTS

2 SIDES · MEASURED LOCAL

B-SIDE · BT1 REDIS SERVER · JAVA github.com/praxstack/redis-server-java RESP2 protocol parser over a single-thread event-loop I/O core with a bespoke hashmap engine — structured for O(1) commands and correct expiry semantics. ▶ SET 154k rps ▶ GET 195k rps INCR 128k rps 42 tests ~78% of C	B-SIDE · BT2 MARKDOWN VIEWER APP github.com/praxstack/markdown-viewer-app Zero-config reader with a plugin architecture — drop-in transforms for diagrams, math, and syntax. Ships tiny and dependency-free. ▶ 155+ Vitest tests < 100KB gz zero deps plugin API
B-SIDE · BT3 OPEN SOURCE · FABRIC github.com/danielmiessler/Fabric Three PRs merged into the 42k-star Go AI-tooling framework — AWS Bedrock bearer-token auth, dynamic region fetching, and a streaming goroutine-deadlock fix. ▶ 3 PRs merged 42k★ upstream Go	

DISCO

DISCOGRAPHY NOTES · EDUCATION & RECOGNITION

ON FILE · CERTIFIED

EDUCATION M.C.A. · CS MNNIT Prayagraj · 2019 — 2022 · GPA 9.25 / 10 B.SC. (HONS) · CS Bhaskaracharya College · DU · 2015 — 2018	RECOGNITION AWARD Amazon Pay · Merchants Categories Champion 2023
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